

Part I

Recitation Questions

Problem 1

February 21, 2026

Problem 1

Problem 1 *Determine whether the following statements are true or false and give either an explanation or a counterexample.*

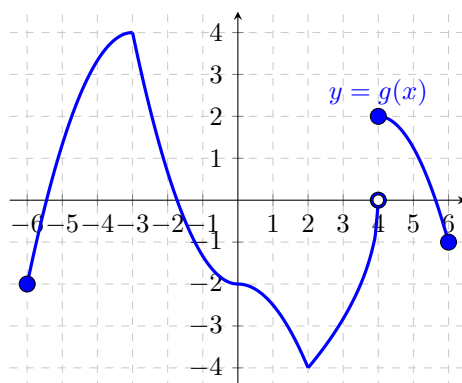
- (a) *The function $f(x) = \sqrt{x}$ has a local maximum on the interval $[0, 1]$.*
- (b) *If $f'(2) = 0$, then $x = 2$ is either a local maximum or local minimum of f .*

Problem 2

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Problem 2

Problem 2 The entire graph of a function g is given below.



Based on the graph of g , answer the questions below.

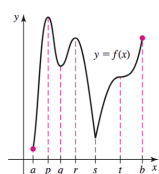
- List the x -coordinates of all critical points of g .
- List the x -coordinates of all critical points of g where $g'(x) = 0$.
- List the x -coordinates of all critical points of g where $g'(x)$ is **undefined**.
- List the x -coordinates of all local maximums of g .
- List the x -coordinates of all local minimums of g .
- List all intervals where g is both decreasing AND concave down.
- List all intervals where g is both decreasing AND concave up.
- List all intervals where g is both increasing AND concave down.
- List all intervals where g is both increasing AND concave up.
- List the x -coordinates of all inflection points of g .

Problem 3

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Problem 3

Problem 3 For each point in the interval (a,b) and identified on the graph below, determine if the function f has a critical point, a local max or min at that point.



Problem 4

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Problem 4

Problem 4 *Locate the critical points and use the second derivative test to determine whether they correspond to local maxima or local minima. **EX-PLAIN.***

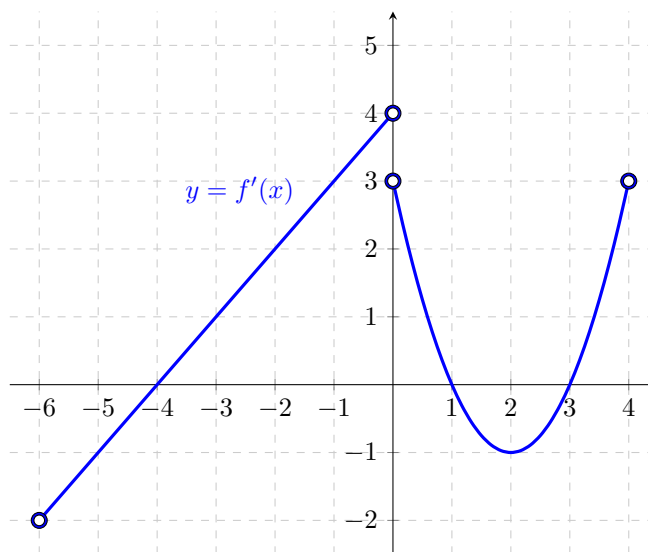
$f(x) = (x + c)^4$ where c is a positive constant

Problem 5

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Problem 5

Problem 5 Let f be a function that is continuous on its domain $(-6, 4)$. The graph of f' , the derivative of f , is given below.



Based on the graph of f' , answer the question below.

- List x-coordinates of all critical points of f .
- List x-coordinates of all critical points of f where $f'(x) = 0$.
- List x-coordinates of all critical points of f where $f'(x)$ is **undefined**.
- List x-coordinates of all local minimums of f .
- List x-coordinates of all local maximums of f .
- List all intervals where f is decreasing and concave down.
- List all intervals where f is decreasing and concave up.
- List all intervals where f is increasing and concave down.
- List all intervals where f is increasing and concave up.
- List x-coordinates of all inflection points of f .

Problem 6

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Problem 6

Problem 6 Find the critical points of f on the given interval. Determine whether the function f has a local maximum, local minimum or no local extremum at each critical point. **EXPLAIN**.

(a) $f(x) = x\sqrt{2-x^2}$ on $(-\sqrt{2}, \sqrt{2})$.

(b) $f(x) = x^3e^{-x}$ on $(-1, 5)$.

(c) $f(x) = x \ln \left(\frac{x}{5} \right)$ on $(0, 5)$.

Problem 7

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Problem 7

Problem 7 Let $f(x) = \frac{1}{1+x^2}$. Find the following for f :

- (a) f' and f''
- (b) Critical points
- (c) Local extrema (and check your answers with both the first and second derivative tests)
- (d) Inflection points. **EXPLAIN**.

Problem 8

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Problem 8

Problem 8 *Sketch a possible graph of a function f that has the following properties:*

- (a) *f is defined on the interval $(0, 6)$.*
- (b) *f has no local maximums.*
- (c) *f has exactly two local minimums.*

Problem 9

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Problem 9

Problem 9 Consider the parabola $f(x) = ax^2 + bx + c$ where a, b, c are constants. For what values of a, b, c is f concave up? For what values of a, b, c is f concave down?